University of Chicago **Spring 2026**
Department of Economics **ECMA 31150**

Problem Set 1

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Please upload your answers on Canvas, with code attached. You are encouraged to collaborate, but please write up your answers independently and do not share code chunks. Use any software you like, although you learn the most by limiting the use of generative AI. If you find any errors or require clarification, please let me know right away.

Question 1

i) Consider the AR(2) process

$$y_t = 2 + \tfrac{5}{6} y_{t-1} - \tfrac{1}{6} y_{t-2} + u_t, \quad u_t \overset{iid}{\sim} N(0, 2).$$

Show that this process is covariance stationary (you may ignore the issue of what the distribution of the initial values needs to be to ensure this). What is the optimal two-step-ahead forecast $y_{t+2|t} \equiv E[y_{t+2} \mid y_t, y_{t-1}, y_{t-2}, \dots]$? What is the associated MSFE $E[(y_{t+2} - y_{t+2|t})^2]$? [Note: Here we assume the coefficients in the model are known to us, not estimated.]

ii) Now consider the AR(3) process

$$y_t = 2 - \tfrac{1}{8} y_{t-3} + u_t, \quad u_t \overset{iid}{\sim} N(0, 4).$$

Repeat the same questions as in part (i) for this model.

Question 2

Let $\{y_t\}$ be a covariance stationary AR(p_y) process:

$$a(L) y_t = u_t, \quad u_t \sim WN(\sigma_u^2), \quad a(L) \equiv 1 - \sum_{\ell=1}^{p_y} a_\ell L^\ell.$$

Let $\{z_t\}$ be a covariance stationary AR(p_z) process:

$$b(L)z_t = v_t, \quad v_t \sim WN(\sigma_v^2), \quad b(L) \equiv 1 - \sum_{\ell=1}^{p_z} b_\ell L^\ell.$$

Assume that $\{u_t\}$ and $\{v_t\}$ are independent processes. Define $w_t \equiv y_t + z_t$ for all t. Show that $\{w_t\}$ is an ARMA(p, q) process, where $p \equiv p_y + p_z$ and $q \equiv \max\{p_y, p_z\}$. [Hint: What does $a(L)b(L)w_t$ equal?]

Question 3

i) Consider a stationary AR(1) model

$$y_t = \rho y_{t-1} + u_t, \quad u_t \overset{iid}{\sim} (0, \sigma_u^2),$$

where $|\rho| < 1$ and $\sigma_u^2 < \infty$. Here the notation "$u_t \overset{iid}{\sim} (0, \sigma_u^2)$" means that u_t is i.i.d. over time with mean zero and variance σ_u^2. Prove that

$$\hat{\delta} \equiv \frac{1}{T} \sum_{t=1}^{T} u_t y_{t-1} \xrightarrow{p} 0 \quad \text{as} \quad T \to \infty.$$

[Hint: First argue that the product $u_t y_{t-1}$ is serially uncorrelated. Then compute the mean and variance of $\hat{\delta}$. Finally, apply Chebyshev's inequality.]

ii) Now consider the MA(1) model

$$y_t = u_t + \theta u_{t-1}, \quad u_t \overset{iid}{\sim} (0, \sigma_u^2),$$

where $\sigma_u^2 < \infty$. Prove that

$$\hat{\mu} \equiv \frac{1}{T} \sum_{t=1}^{T} y_t \xrightarrow{p} 0 \quad \text{as} \quad T \to \infty.$$

[Hint: The argument will have to be slightly different from before, since y_t is no longer serially uncorrelated.]

Question 4

Go to the St. Louis Fed's FRED website and download the IMF's Global Price Index of All Commodities (series code: PALLFNFINDEXM). For all the statistical analysis below, only use data until (and including) December 2025. Transform the data to logged first differences prior to the following analysis.

i) Plot the resulting time series.

ii) Test whether the series is white noise, based on an AR(12) regression. Allow for heteroskedastic shocks.

iii) Plot the BIC and AIC criteria for AR(p) models of order $p = 1, 2, \ldots, 24$. Which order is selected by the two criteria?

iv) Using the AIC-selected lag order, estimate an AR model and report heteroskedasticity-robust standard errors for the coefficients. Also report an estimate of the innovation variance.

v) Now assuming an AR(1) model, forecast the values of the series in January and February of 2026. Since we actually do observe data for these months, you can compute the forecast errors of the two predictions – report them.

vi) Now assuming an AR(1) model with i.i.d. normally distributed shocks, report a 90% forecast interval for the series in February 2026 (two periods ahead). Does the interval contain the actual realized value? [Note: we did not explicitly discuss in lecture how to form multi-step forecast intervals, but you should be able to work out the logic.]